Microscreens

Why use them?

and

How do they work?

WHAT WILL WE COVER

What is a microscreen and what does it do?

Cellulose entering a WWTP:

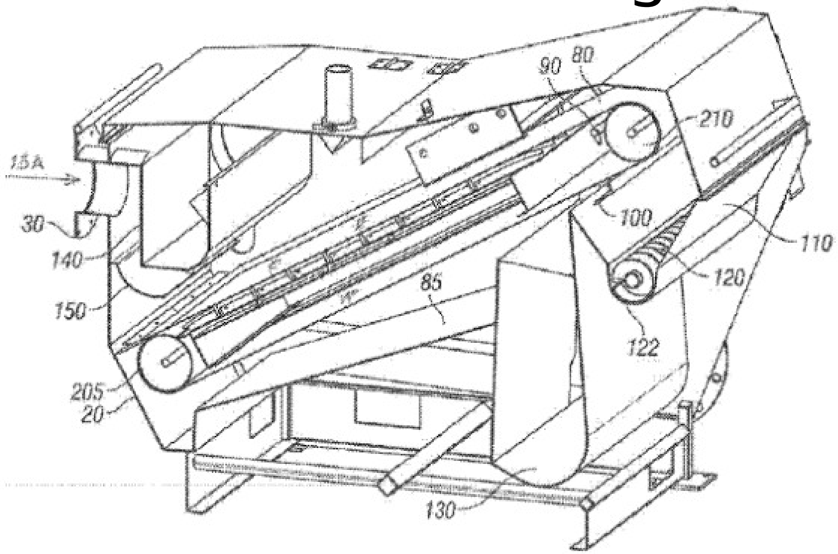
- Where does it comes from?
- Where does it go?

How do microscreens fit into Flagstaff’s needs.

Microscreen & Landfill

Solids Removal
& Dewatering

Raw
Sewage



Damp
biosolids to
Landfill

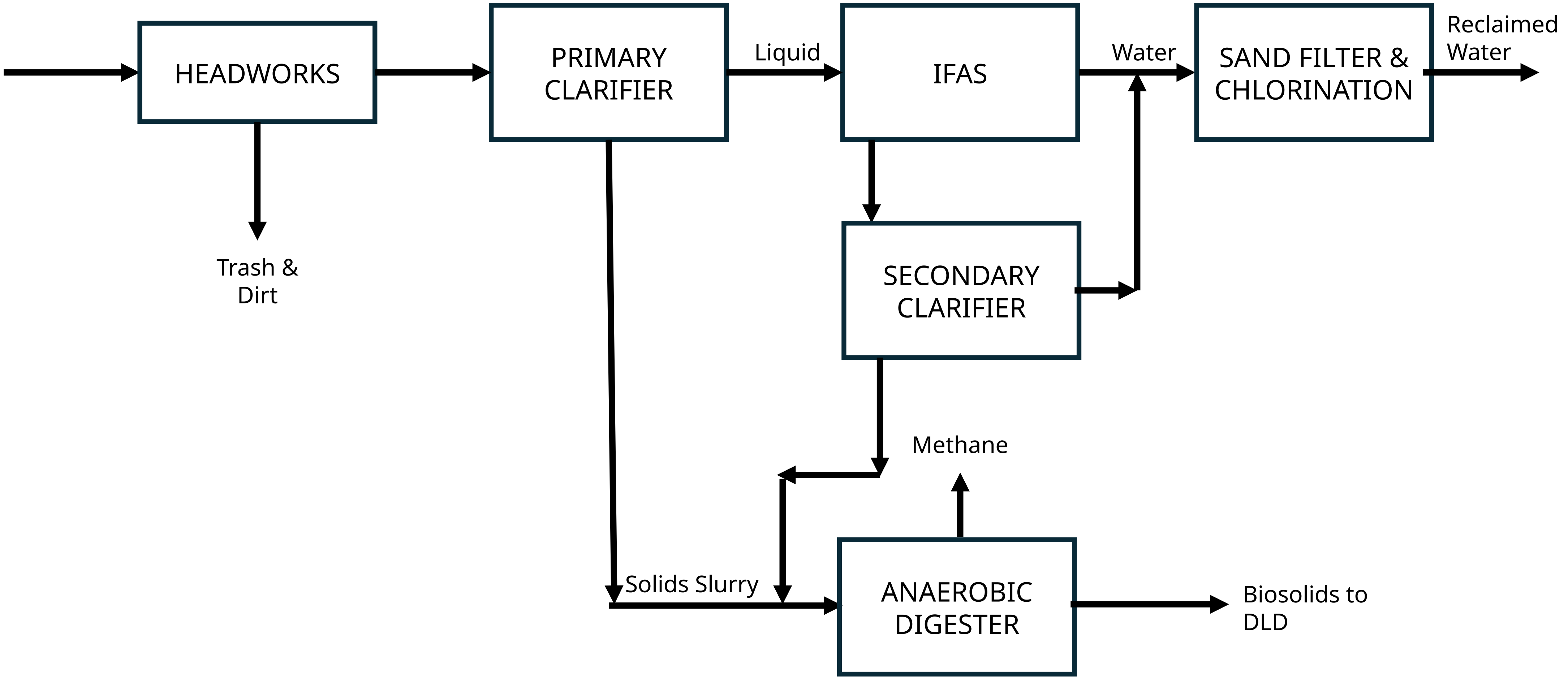
Microscreen

Why the Microscreen?

The microscreen removes solids that the primary settlers can not.

- About 95% removal of cellulose and lignin
- While the primary settler today
- Primary settler captures about 80%
    - Depends on gravity
    - Some cellulose density is close to water

WWTP Essentials



Where is the Microscreen Used

1. Existing WWTPs with solids overloading
   - Reduces backend biosolids
   - Unloads all the processes in the plant
   - Can help remove anaerobic digesters
   - Saves energy

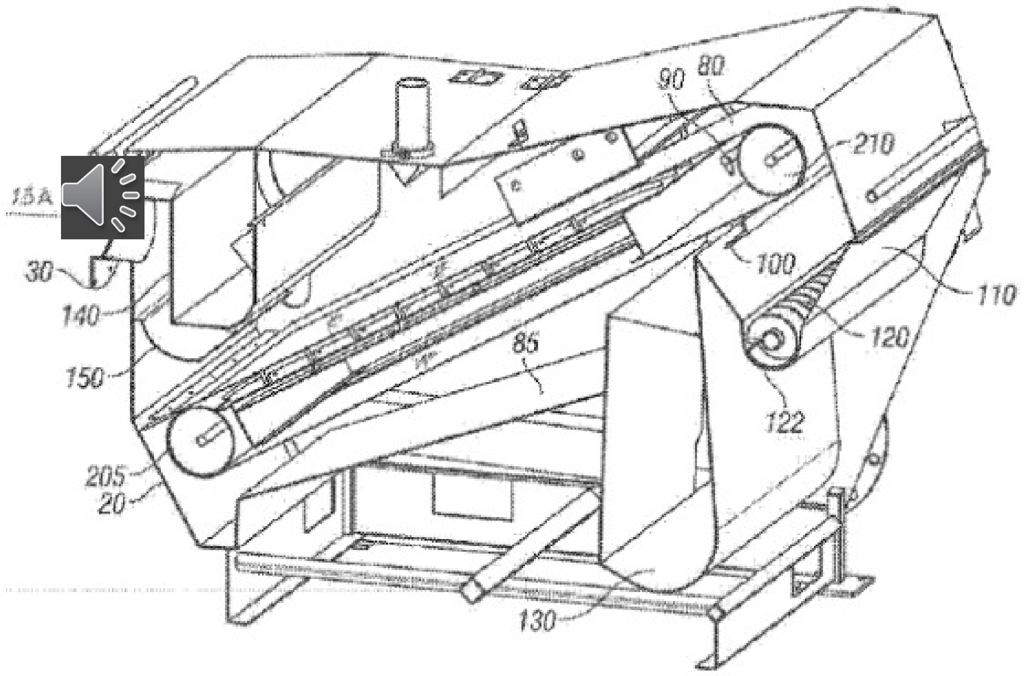
2. New WWTPs
   - Replaces need for primary clarifiers
   - Can relocate units at existing plants
   - Can add additional units when needed

Microscreen on Raw Sewage
(AKA “Rotating Belt Filter” – RBF)

Removes about ½ of TSS solids from raw sewage

Solids are largely fibrous (easily removed)

Solids produced are drier
(35% solids versus 20% for digested solids) – less than half the water

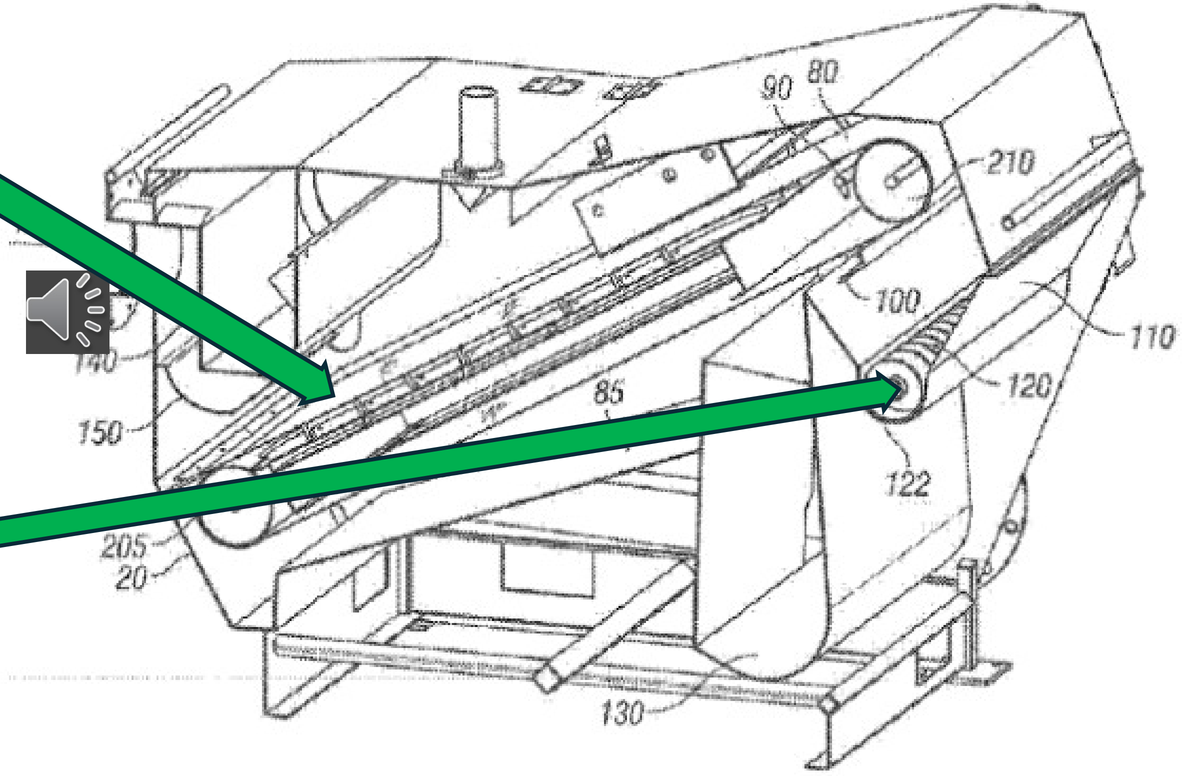


Microscreen Components

Consists of two components

Belt

Dewatering Auger Press



Auger Press serves the same function at the moving-plate auger in the SHINCCI dryer application

Microscreen Principles

Selectively pull cellulose from raw sewage

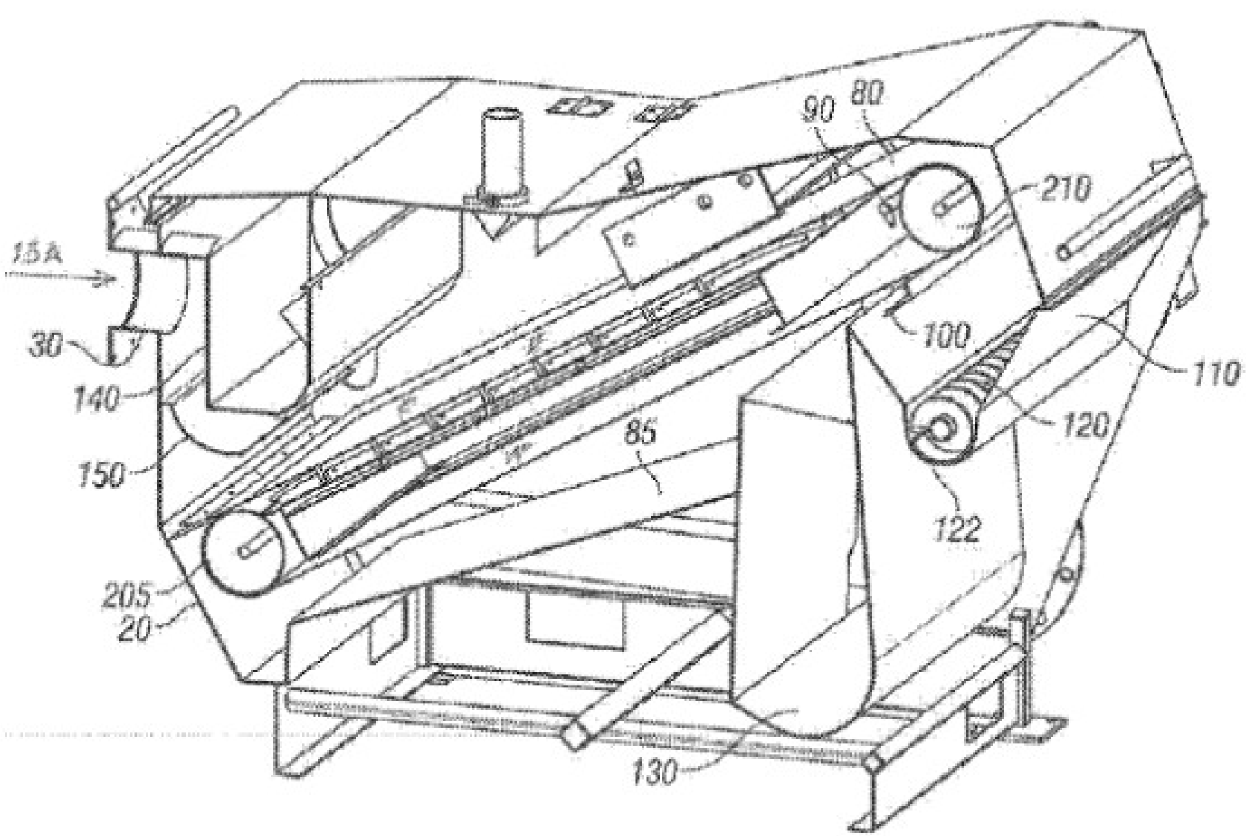
Screens capture cellulose fibers while letting most bacteria sized particles through.

But,

A “mat” buildup on the screen is possible to capture smaller particles.

Microscreen Demo

Microscreen moc



Microscreen Principles

Using Screen Boxes and a Squeeze Rollers to Simulate the Microscreen

Screen and Particle Sizes:

Screen Sizes in Demo

40 mesh has holes of about 450 Microns

100 mesh has holes of about 150-200 microns

Microscreen Principles

Screen and Particle Sizes:

Screen

40 mesh has holes of about 450 Microns

100 mesh has holes of about 150-200 microns

Sewage

Toilet Paper fibers are 500 to 4,500 microns in length, with diameters in the 10 to 50-micron range

Bacteria are less than 5 microns in length with most under 2 micron.

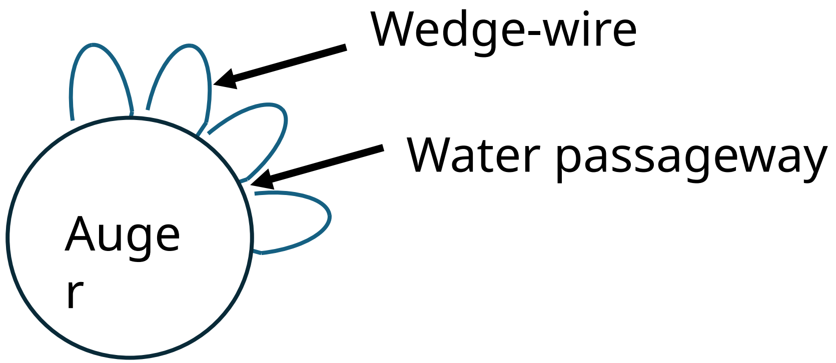
Diatomaceous Earth (DE) (simulates bacteria) size is 10-50 micron.

Microscreen Demo

Auger Performance

Real Microscreen uses
auger press
with wedge-wire

End View of Auger



Demo will use roller to simulate the action of the auger.

Microscreen Performance

Removes nearly all cellulose from incoming sewage

- Cellulose is about 30% of incoming solids
- Studies show Microscreen captures at least 95-99% of fibers
- For comparison, primary settler captures about 75%, rest goes to IFAS
- Removing the cellulose gets about 50% of the incoming solids

Can capture more solids by allowing mat buildup

- Belt speed gives great control over Microscreen performance
- Can be automated to always remove more solids up to Microscreen capacity

Cellulose removal improves performance of all downstream units

Microscreen

- Questions?